

CockroachDB High-Availability Lab

3-Node CockroachDB + Dual HAProxy + Keepalived Floating VIP

Final validated implementation, commands, behaviors, failover testing, and root-cause analysis

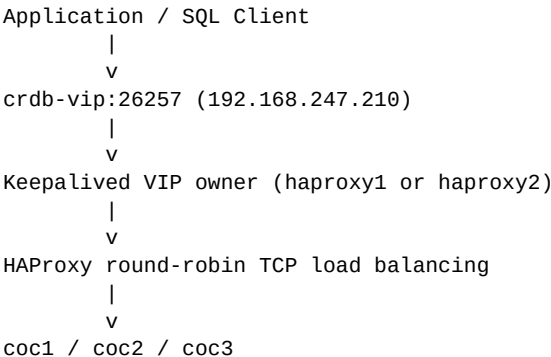
1. Executive Summary

This document records the completed self-hosted CockroachDB high-availability lab. The database tier consists of three CockroachDB nodes. Two HAProxy servers provide SQL load balancing, and Keepalived provides a single floating virtual IP address (VIP) for application connectivity. The final design was validated by stopping the primary HAProxy service, confirming VIP movement to the backup server, running SQL successfully through the VIP, restarting the primary HAProxy service, and confirming automatic failback.

2. Final Architecture

Role	Hostname	IP Address	Purpose
CockroachDB node 1	coc1	192.168.247.156	Database node; SQL 26257, DB Console/health 8080
CockroachDB node 2	coc2	192.168.247.198	Database node; SQL 26257, DB Console/health 8080
CockroachDB node 3	coc3	192.168.247.199	Database node; SQL 26257, DB Console/health 8080
Primary load balancer	haproxy1	192.168.247.200	Preferred Keepalived MASTER; HAProxy SQL endpoint
Backup load balancer	haproxy2	192.168.247.202	Keepalived BACKUP; takes VIP during failure
Floating application endpoint	crdb-vip	192.168.247.210	Single application SQL endpoint on port 26257

Normal connection path:



3. Completed Components and Versions

- RHEL 9.6 on all nodes.
- CockroachDB v26.2.3 on the three database nodes.
- HAProxy 2.8.14 on both load balancer servers.

- Keepalived 2.2.8 on both load balancer servers.
- CockroachDB cluster running in insecure lab mode.
- SELinux remains Enforcing.
- firewalld remains enabled.
- HAProxy statistics page available on TCP port 8404 on each HAProxy server.

4. Network and Hostname Configuration

4.1 Static IP configuration for haproxy1

```
nmcli connection modify ens160 \
  ipv4.method manual \
  ipv4.addresses 192.168.247.200/24 \
  ipv4.gateway 192.168.247.2 \
  ipv4.dns 192.168.247.2

nmcli connection up ens160

ip -4 addr show ens160
ip route
```

Command	Behavior / Expected Result
nmcli connection modify ... ipv4.method manual	Converts the connection profile from DHCP to a persistent static IPv4 configuration.
Command	Behavior / Expected Result
nmcli connection up ens160	Reactivates the connection profile. SSH can briefly disconnect while the profile is reapplied.
Command	Behavior / Expected Result
valid_lft forever	Confirms the address is static rather than DHCP-leased.

4.2 Static IP configuration for haproxy2

```
nmcli connection modify ens160 \
  ipv4.method manual \
  ipv4.addresses 192.168.247.202/24 \
  ipv4.gateway 192.168.247.2 \
  ipv4.dns 192.168.247.2

nmcli connection up ens160
```

4.3 /etc/hosts entries

The following entries were added to haproxy1, haproxy2, coc1, coc2, and coc3:

```
192.168.247.200 haproxy1
192.168.247.202 haproxy2
192.168.247.210 crdb-vip
```

Command	Behavior / Expected Result
getent ahostsv4 haproxy1 haproxy2 crdb-vip	Validates IPv4 hostname resolution. This avoids confusion when the local hostname also resolves to a link-local IPv6 address.

5. HAProxy Configuration

The same HAProxy configuration is used on haproxy1 and haproxy2:

```
global
  maxconn 4096
  user haproxy
```

```

group haproxy
daemon

defaults
mode tcp
timeout connect 10s
timeout client 1h
timeout server 1h
timeout check 5s
option clitcpka

listen cockroach_sql
bind 0.0.0.0:26257
mode tcp
balance roundrobin
option httpchk GET /health?ready=1

server coc1 192.168.247.156:26257 check port 8080
server coc2 192.168.247.198:26257 check port 8080
server coc3 192.168.247.199:26257 check port 8080

listen haproxy_stats
bind 0.0.0.0:8404
mode http
stats enable
stats uri /stats
stats refresh 5s

```

5.1 HAProxy command behavior

Command	Behavior / Expected Result
<code>haproxy -c -f /etc/haproxy/haproxy.cfg</code>	Performs a syntax check without starting or reloading HAProxy.
Command	Behavior / Expected Result
<code>systemctl enable --now haproxy</code>	Starts HAProxy immediately and enables it at boot.
Command	Behavior / Expected Result
<code>systemctl is-active haproxy</code>	Returns active when the service is running.
Command	Behavior / Expected Result
<code>ss -lntp grep -E ":26257 :8404"</code>	Confirms HAProxy is listening for SQL and statistics traffic.
Command	Behavior / Expected Result
<code>curl http://127.0.0.1:8404/stats</code>	Displays the HAProxy statistics page locally.

5.2 HAProxy health-check behavior

HAProxy sends an HTTP GET request to each CockroachDB node on port 8080 using `/health?ready=1`. The SQL connection itself is forwarded to port 26257. A backend is marked UP only when the CockroachDB readiness endpoint returns HTTP 200.

6. SELinux and Firewall Configuration

6.1 HAProxy SELinux port labels

```

semanage port -a -t http_port_t -p tcp 26257
semanage port -a -t http_port_t -p tcp 8404

semanage port -l | grep http_port_t

```

If a port is already defined, use `-m` instead of `-a` to modify the existing mapping.

6.2 firewall rules

```

firewall-cmd --permanent --add-port=26257/tcp
firewall-cmd --permanent --add-port=8404/tcp

```

```

firewall-cmd --permanent --add-protocol=vrrp
firewall-cmd --reload

firewall-cmd --list-ports
firewall-cmd --list-protocols

```

Command	Behavior / Expected Result
--add-protocol=vrrp	Allows IP protocol 112, used by Keepalived VRRP advertisements. VRRP is not a TCP or UDP port.

7. Keepalived Configuration

7.1 Final HAProxy health-check script

Final file on both HAProxy servers: /usr/local/bin/check_haproxy.sh

```

#!/bin/bash

PIDFILE=/run/haproxy.pid

[ -r "$PIDFILE" ] || exit 1

PID=$(cat "$PIDFILE")

[ -n "$PID" ] || exit 1

kill -0 "$PID" 2>/dev/null

```

Command	Behavior / Expected Result
[-r "\$PIDFILE"]	Fails if the HAProxy PID file does not exist or cannot be read.
Command	Behavior / Expected Result
PID=\$(cat "\$PIDFILE")	Reads the HAProxy process ID from /run/haproxy.pid.
Command	Behavior / Expected Result
kill -0 "\$PID"	Does not terminate the process. It checks whether the process exists and is accessible. Exit 0 means healthy; nonzero means failed.

7.2 haproxy1 Keepalived configuration

```

global_defs {
    router_id HAPROXY1
    enable_script_security
    script_user root
}

vrrp_script check_haproxy {
    script "/usr/local/bin/check_haproxy.sh"
    interval 2
    timeout 2
    fall 2
    rise 2
    weight -20
}

vrrp_instance VI_CRDB {
    state MASTER
    interface ens160
    virtual_router_id 51
    priority 110
    advert_int 1

    unicast_src_ip 192.168.247.200
    unicast_peer {
        192.168.247.202
    }
}

```

```

}

authentication {
    auth_type PASS
    auth_pass CRDBHA01
}

virtual_ipaddress {
    192.168.247.210/24 dev ens160
}

track_script {
    check_haproxy
}
}

```

7.3 haproxy2 Keepalived configuration

```

global_defs {
    router_id HAPROXY2
    enable_script_security
    script_user root
}

vrrp_script check_haproxy {
    script "/usr/local/bin/check_haproxy.sh"
    interval 2
    timeout 2
    fall 2
    rise 2
    weight -20
}

vrrp_instance VI_CRDB {
    state BACKUP
    interface ens160
    virtual_router_id 51
    priority 100
    advert_int 1

    unicast_src_ip 192.168.247.202
    unicast_peer {
        192.168.247.200
    }

    authentication {
        auth_type PASS
        auth_pass CRDBHA01
    }

    virtual_ipaddress {
        192.168.247.210/24 dev ens160
    }

    track_script {
        check_haproxy
    }
}

```

7.4 Keepalived setting behavior

Command	Behavior / Expected Result
state MASTER / BACKUP	Defines the initial role preference. Election is still based on effective priority.
Command	Behavior / Expected Result
priority 110 / 100	haproxy1 normally owns the VIP because 110 is higher than

Command	Behavior / Expected Result
	100.
Command	Behavior / Expected Result
weight -20	A failed health check subtracts 20 from effective priority. haproxy1 becomes 90 and haproxy2 becomes 80 when unhealthy.
Command	Behavior / Expected Result
interval 2, fall 2	The script runs every 2 seconds. Two consecutive failures are required before the state is considered failed, producing roughly a 4-6 second failover in this lab.
Command	Behavior / Expected Result
rise 2	Two consecutive successful checks are required to recover normal priority.
Command	Behavior / Expected Result
unicast_peer	VRRP advertisements are sent directly between the two load balancers instead of using multicast.
Command	Behavior / Expected Result
virtual_router_id 51	Both servers must use the same value to participate in the same VRRP instance.
Command	Behavior / Expected Result
advert_int 1	VRRP advertisements are sent every second.

8. Custom SELinux Policy for Keepalived Health Check

SELinux correctly prevented the keepalived_t domain from reading the HAProxy PID file. A narrow local policy was installed on both HAProxy servers to permit only the required file access.

```
module keepalived_haproxy_pid 1.0;

require {
    type keepalived_t;
    type haproxy_var_run_t;
    class file { getattr open read };
}

allow keepalived_t haproxy_var_run_t:file { getattr open read };

checkmodule -M -m \
  -o /root/keepalived_haproxy_pid.mod \
  /root/keepalived_haproxy_pid.te

semodule_package \
  -o /root/keepalived_haproxy_pid.pp \
  -m /root/keepalived_haproxy_pid.mod

semodule -i /root/keepalived_haproxy_pid.pp

systemctl restart keepalived
```

Command	Behavior / Expected Result
checkmodule	Compiles the human-readable SELinux type-enforcement source into a module object.
Command	Behavior / Expected Result
semodule_package	Packages the module into an installable .pp policy file.
Command	Behavior / Expected Result
semodule -i	Installs the local SELinux policy persistently.

9. Validation and Operational Commands

Command	Behavior / Expected Result
keepalived --config-test	Validates /etc/keepalived/keepalived.conf. No output indicates success.
Command	Behavior / Expected Result
systemctl status keepalived --no-pager -l	Shows service state and recent VRRP transitions.
Command	Behavior / Expected Result
journalctl -u keepalived --since "5 minutes ago" --no-pager	Shows detailed VIP election, health-check, priority, and failover events.
Command	Behavior / Expected Result
ip -4 addr show ens160 grep 192.168.247.210	Shows whether the local server currently owns the VIP.
Command	Behavior / Expected Result
ausearch -m AVC -ts recent	Displays recent SELinux access denials.
Command	Behavior / Expected Result
ls -lZ /run/haproxy.pid	Displays Linux permissions and SELinux context for the HAProxy PID file.
Command	Behavior / Expected Result
firewall-cmd --list-protocols	Confirms vrrp is allowed.

10. SQL Validation Through the VIP

The CockroachDB CLI is not required on HAProxy servers. SQL tests were run from coc1, where the CLI was already installed.

```
cockroach sql \
--insecure \
--host=crdb-vip:26257 \
--database=labdb \
--execute="SELECT * FROM employees;"

cockroach sql \
--insecure \
--host=crdb-vip:26257 \
--database=labdb \
--execute="SELECT count(*) AS employee_count FROM employees;"
```

Validated result: employee_count = 5 before failover, during failover, and after failback.

11. HAProxy Failover and Failback Test

11.1 Failure simulation

```
# On haproxy1
systemctl stop haproxy
```

Behavior: the local health script fails. After two failed checks, haproxy1 effective priority drops from 110 to 90. haproxy2 remains healthy at priority 100, wins the election, and adds 192.168.247.210 to ens160.

```
# On haproxy2
ip -4 addr show ens160 | grep 192.168.247.210
```

11.2 SQL continuity test

```
# On coc1
cockroach sql \
--insecure \
--host=crdb-vip:26257 \
--database=labdb \
--execute="SELECT count(*) AS employee_count FROM employees;"
```

Observed result: query succeeded and returned 5 rows while haproxy2 owned the VIP.

11.3 Recovery and failback

```
# On haproxy1
systemctl start haproxy
systemctl is-active haproxy

# After approximately 6 seconds
ip -4 addr show ens160 | grep 192.168.247.210
```

Behavior: the health check succeeds twice, haproxy1 returns to priority 110, preempts haproxy2, and the VIP moves back to haproxy1. A final SQL query through crdb-vip succeeded with employee_count = 5.

12. Normal Operating State

Component	Expected State	Verification
haproxy1	HAProxy active; Keepalived MASTER; owns VIP	systemctl is-active haproxy keepalived; ip addr
haproxy2	HAProxy active; Keepalived BACKUP; no VIP	systemctl is-active haproxy keepalived; ip addr
VIP	192.168.247.210 reachable on SQL port 26257	nc -zv crdb-vip 26257 or CockroachDB SQL query
CockroachDB backends	All three shown UP in HAProxy stats	HAProxy stats page on port 8404

13. Root Cause Analysis and Resolutions

This section summarizes the major issues encountered, the evidence used to isolate each cause, and the permanent resolution.

13.1 CockroachDB systemd service failed with YAML unmarshal error

Root cause	The service used --log=file-defaults:dir=/var/log/cockroach, which was interpreted as structured logging YAML and failed parsing.
Resolution	Replaced the argument with --log-dir=/var/log/cockroach.
Behavior after fix	CockroachDB services started normally and wrote logs to the intended directory.
Key evidence	Commands used included systemctl status, journalctl -u keepalived, ausearch -m AVC -ts recent, ls -lZ, ip addr, and SQL queries through crdb-vip.

13.2 HAProxy could not bind to ports 26257 and 8404 while SELinux was Enforcing

Root cause	The ports were not labeled with an SELinux type permitted for the HAProxy domain.
Resolution	Added both ports to http_port_t using semanage port.
Behavior after fix	HAProxy started successfully while SELinux remained Enforcing.
Key evidence	Commands used included systemctl status, journalctl -u keepalived, ausearch -m AVC -ts recent, ls -lZ, ip addr, and SQL queries through crdb-vip.

13.3 Keepalived health script returned status 126

Root cause	The original script called /usr/bin/systemctl. AVC logs showed keepalived_t was denied execute/getattr access to systemd_systemctl_exec_t.
Resolution	Replaced the systemctl-based check with a PID-file and kill -0 process existence check.
Behavior after fix	The script became executable under Keepalived, but initially

	returned status 1 because PID-file access was still denied.
Key evidence	Commands used included systemctl status, journalctl -u keepalived, ausearch -m AVC -ts recent, ls -lZ, ip addr, and SQL queries through crdb-vip.

13.4 Keepalived health script returned status 1 after PID-based rewrite

Root cause	AVC logs showed keepalived_t was denied read access to /run/haproxy.pid labeled haproxy_var_run_t.
Resolution	Installed a minimal local SELinux module allowing keepalived_t to getattr, open, and read haproxy_var_run_t files.
Behavior after fix	VRRP_Script(check_haproxy) succeeded on both HAProxy servers and effective priorities returned to 110 and 100.
Key evidence	Commands used included systemctl status, journalctl -u keepalived, ausearch -m AVC -ts recent, ls -lZ, ip addr, and SQL queries through crdb-vip.

13.5 haproxy2 initially advertised effective priority 80

Root cause	Its base priority was 100 and the failed track script applied weight -20.
Resolution	Fixed the health script and SELinux policy.
Behavior after fix	haproxy2 advertised normal priority 100 and remained BACKUP while haproxy1 was healthy.
Key evidence	Commands used included systemctl status, journalctl -u keepalived, ausearch -m AVC -ts recent, ls -lZ, ip addr, and SQL queries through crdb-vip.

13.6 haproxy1 temporarily lost MASTER role after haproxy2 health check was fixed

Root cause	haproxy1 still had a failed health script, reducing its priority from 110 to 90. Healthy haproxy2 remained at 100 and won the election.
Resolution	Applied the same PID-based script and SELinux policy on haproxy1.
Behavior after fix	haproxy1 recovered to priority 110 and correctly became MASTER.
Key evidence	Commands used included systemctl status, journalctl -u keepalived, ausearch -m AVC -ts recent, ls -lZ, ip addr, and SQL queries through crdb-vip.

13.7 CockroachDB CLI command failed on HAProxy server

Root cause	The CockroachDB client binary was never installed on the load balancer nodes, and it is not required for HAProxy operation.
Resolution	Ran SQL validation from coc1, where the CockroachDB CLI already existed.
Behavior after fix	VIP connectivity was validated without adding unnecessary software to HAProxy servers.
Key evidence	Commands used included systemctl status, journalctl -u keepalived, ausearch -m AVC -ts recent, ls -lZ, ip addr, and SQL queries through crdb-vip.

13.8 getent hosts haproxy1 returned a link-local IPv6 address

Root cause	The local machine hostname also resolved through the system hostname mechanism, which was preferred by getent hosts.
Resolution	Validated explicit IPv4 mappings with getent ahostsv4 and retained correct /etc/hosts entries.
Behavior after fix	IPv4 name resolution for haproxy1, haproxy2, and crdb-vip was confirmed.
Key evidence	Commands used included systemctl status, journalctl -u keepalived, ausearch -m AVC -ts recent, ls -lZ, ip addr, and SQL queries through crdb-vip.

13.9 VIP appeared to remain on haproxy1 immediately after HAProxy was stopped

Root cause	The Keepalived configuration requires two failed checks at 2-second intervals before reducing priority, plus VRRP election time.
Resolution	Waited approximately 6 seconds before checking VIP ownership on haproxy2.
Behavior after fix	VIP movement occurred as designed and SQL remained available.
Key evidence	Commands used included systemctl status, journalctl -u keepalived, ausearch -m AVC -ts recent, ls -lZ, ip addr, and SQL queries through crdb-vip.

14. Final Application Endpoint

Applications and client tools should use the floating endpoint, not an individual HAProxy server:

Host: crdb-vip
IP: 192.168.247.210
Port: 26257

In this lab the CockroachDB cluster is insecure, so the test CLI uses `--insecure`. For a production deployment, configure CockroachDB TLS certificates and require secure client connections.